

Ex. No.: 11b)

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LRU

Aim:

To write a c program to implement LRU page replacement algorithm.

Algorithm:

- 1: Start the process
- 2: Declare the size
- 3: Get the number of pages to be inserted
- 4: Get the value
- 5: Declare counter and stack
- 6: Select the least recently used page by counter value
- 7: Stack them according to the selection.
- 8: Display the values
- 9: Stop the process

Program Code:

```
#include <stdio.h>

int main () {
    int frames, pages, i, j, k, flag, counter=0, min, page_faults=0;
    int page[50], frame[10], used[10];
    printf ("Enter number of frames: ");
    scanf ("%d", &frames);
    printf ("Enter number of pages: ");
    scanf ("%d", &pages);
    printf ("Enter the page reference string: ");
    for (i=0; i<pages; i++) {
        scanf ("%d", &page[i]);
    }
    for (i=0; i<frames; i++) {
        frame[i] = -1;
        used[i] = 0;
    }
}
```



```

for (i=0; i < pages; i++) {
    flag = 0;
    for (j=0; j < frames; j++) {
        if (frame[j] == page[i]) {
            used[j] = count++;
            flag = 1;
            break;
        }
    }
}

```

```

if (flag == 0) {
    int pos = 0;
    if (i < frames) { pos = i; }
    else { min = used[0];
        for (j=1; j < frames; j++) {
            if (used[j] < min) {
                min = used[j];
                pos = j;
            }
        }
    }
}

```

```

frame[pos] = page[i];
used[pos] = count++;
page-faults++;

```

```

printf("Frame after page %d: ", page[i]);
for (k=0; k < frames; k++) {
    if (frame[k] != -1)
        printf("%d", frame[k]);
    else printf("- ");
}

```

```

printf("\n");

```

```

printf("Total page Faults: %d\n", page-faults);

```

```

return 0;
}

```


INPUT:

Enter number of frames: 3

Enter number of pages: 12

Enter the page reference string: 1 2 3 4 1 2 5 1 2 3 4 5

OUTPUT:

Frame after page 1: 1 - -

Frame after page 2: 1 2 -

Frame after page 3: 1 2 3

Frame after page 4: 2 3 4

Frame after page 1: 3 4 1

Frame after page 2: 4 1 2

Frame after page 5: 1 2 5

Frame after page 1: 2 5 1

Frame after page 2: 5 1 2

Frame after page 3: 1 2 3

Frame after page 4: 2 3 4

Frame after page 5: 3 4 5

Total page faults: 10

Sample Output :

Enter number of frames: 3

Enter number of pages: 6

Enter reference string: 5 7 5 6 7 3

5 -1 -1

5 7 -1

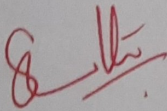
5 7 -1

5 7 6

5 7 6

3 7 6

Total Page Faults = 4



Result:

Thus the code to implement LRU
has been executed successfully